

POWER SUPPLY HUB With Battery Charger, Quad-USB And Variable Outputs

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This article presents a simple, universal low-cost power supply hub with quad-USB and variable voltage outputs, plus a battery charger. There are electronic equipment like battery chargers, electronic lamps and MP3 players that take power from USB ports, but there may not be sufficient ports or sufficient power available from the USB ports on a computer. In the simplest case, we use linear voltage regulators and not switched regulators.

Fig. 1 shows one possible implementation of a power supply hub with the following main characteristics:

1. 9V to 12V AC/DC input
2. Four 5.1V outputs with USB connectors
3. One variable output from 1.2V to 6.5V
4. A constant-current output with a minimum of 10mA and a maximum of 400mA
5. Individual on/off switches (S1 through S6) or jumpers for each output; all outputs controlled by adjustable linear voltage regulators LM317T in TO-220 package

Circuit and working

LM317T is a low-cost, adjustable voltage regulator. It can dissipate up to 20W as heat and produce up to 1.5A of current (refer the datasheet). Six LM317T regulators are used in this project to build a useful power supply hub that provides power supply for four USB devices (USB1 through USB4) at connectors CON3 through CON6.

In addition, one variable output (1.25V to 6.5V) is available at CON7 and a current generator output at CON8 for charging and maintaining

TABLE 1: CURRENT PRODUCED BY IC6					
Closed switch	None	S7	S8	S9	S10
Active resistor	R18=130-ohm	R17=24-ohm	R16=13-ohm	R15=6.2-ohm	R14=3.3-ohm
Approximate output current	9.6mA	52mA*	96mA*	201mA*	379mA*

Note (*) - Current through R18 is neglected

the rechargeable batteries. Connector CON9 in parallel with CON8 is provided as an extra but optional outlet.

Voltages across the battery charger available at CON8 and adjustable output from IC5 available at CON7 should be measured with a voltmeter. Outputs at the USB connectors (CON3 through CON6) have LEDs for visual indications. Glowing of the LEDs indicates the presence of the power supply at each USB connector.

Switches. The circuit has ten switches, S1 through S10. S1 through S5 are used for switching on/off IC1 through IC5 and corresponding V_{OUT1} through V_{OUT5} , respectively.

S6 is the on/off switch for IC6 and for output currents for charging and maintaining rechargeable batteries at CON8 and CON9. S7 through S10 determine the output current produced by IC6.

Voltage and current calculation.

Voltage (V_{OUT1}) produced from IC1 can be calculated with the formula:

$$V_{OUT1} = 1.25V \times (1 + R3/R2) = 1.25V \times 4.09 = 5.11V$$

Similarly, you can calculate voltages V_{OUT2} , V_{OUT3} and V_{OUT4} .

These four voltages should be in the range of 5.1V to 5.2V or near the maximum of 5.25V for the USB standard voltage.

Maximum voltage produced by IC5 can be calculated using the relationship:

PARTS LIST

Semiconductors:

- IC1-IC6 - LM317/LM317T adjustable voltage regulator
- D1-D4 - 1N5404 rectifier diode
- D5-D15 - 1N4007 rectifier diode
- LED1-LED7 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R7-R9, R19-R21 - 3.3-kilo-ohm
- R2, R4, R10, R12 - 220-ohm
- R3, R5, R11, R13 - 680-ohm
- R6 - 240-ohm
- R14 - 3.3-ohm
- R15 - 6.2-ohm
- R16 - 13-ohm
- R17 - 24-ohm
- R18 - 130-ohm
- VR1 - 1-kilo-ohm potentiometer

Capacitors:

- C1 - 10n ceramic disk
- C2, C3 - 4700 μ F, 35V electrolytic
- C4-C12 - 100nF ceramic disk
- C16-C24 - 100nF ceramic disk
- C13-C15, C25, C26 - 100 μ F, 25V electrolytic

Miscellaneous:

- USB1-USB4 - Type-A USB connector
- CON1, CON2, CON7-CON9 - 2-pin terminal connector
- CON3-CON6 - PCB mounted USB connector
- S1-S10 - On/off switch
- F1, F2 - 3A fuse with holder
- X1 (Not shown in circuit) - 230V AC primary to 9V AC/12V AC, 3A secondary transformer or 9V DC to 12V DC, 3A adaptor or 9V-12V battery
- Heat-sink for IC1 through IC6

$$V_{OUT(max)} = 1.25V \times (1 + VR1/R6) = 1.25V \times (1 + 1000\text{-ohm}/240\text{-ohm}) = 6.46V$$

For $V_{OUT(min)}$ you have around 1.25V from the datasheet of LM317T.

Minimum output current produced by IC6 is calculated using the relationship: